SingleRAN

SRAN Networking and Evolution Overview Feature Parameter Description

Issue Draft B

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2025-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added the MBTS that supports Relay in figure "Relationships among typical RATs, NEs, and products on a SingleRAN network" and a product variant of co-MPT multi-RAT base stations. For details, see: ● **4 SingleRAN Solution Overview** ● **4.3.2.2 Co-MPT Multimode Base Station** ● **4.3 Base Station Products**	None	5900 series base stations (macro base stations)

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

3 Overview

Operators are facing the long-term coexistence of multiple RATs in the MBB era. Huawei innovatively proposed the SingleRAN solution in 2008 to help address this challenge. Today, SingleRAN has become a multimode network construction standard in the industry.

The SingleRAN solution uses a unified platform architecture and the software-definable design to implement GSM/UMTS/LTE/NR network coordination and sharing, improving network resource utilization. This solution provides users with ubiquitous broadband service experience.

The SingleRAN solution mainly includes the following products:

- OSS products
- Base station controller products
- Base station products
- Other products

For details, see **4 SingleRAN Solution Overview**.

The following solutions are derived from the evolution of SingleRAN networks:

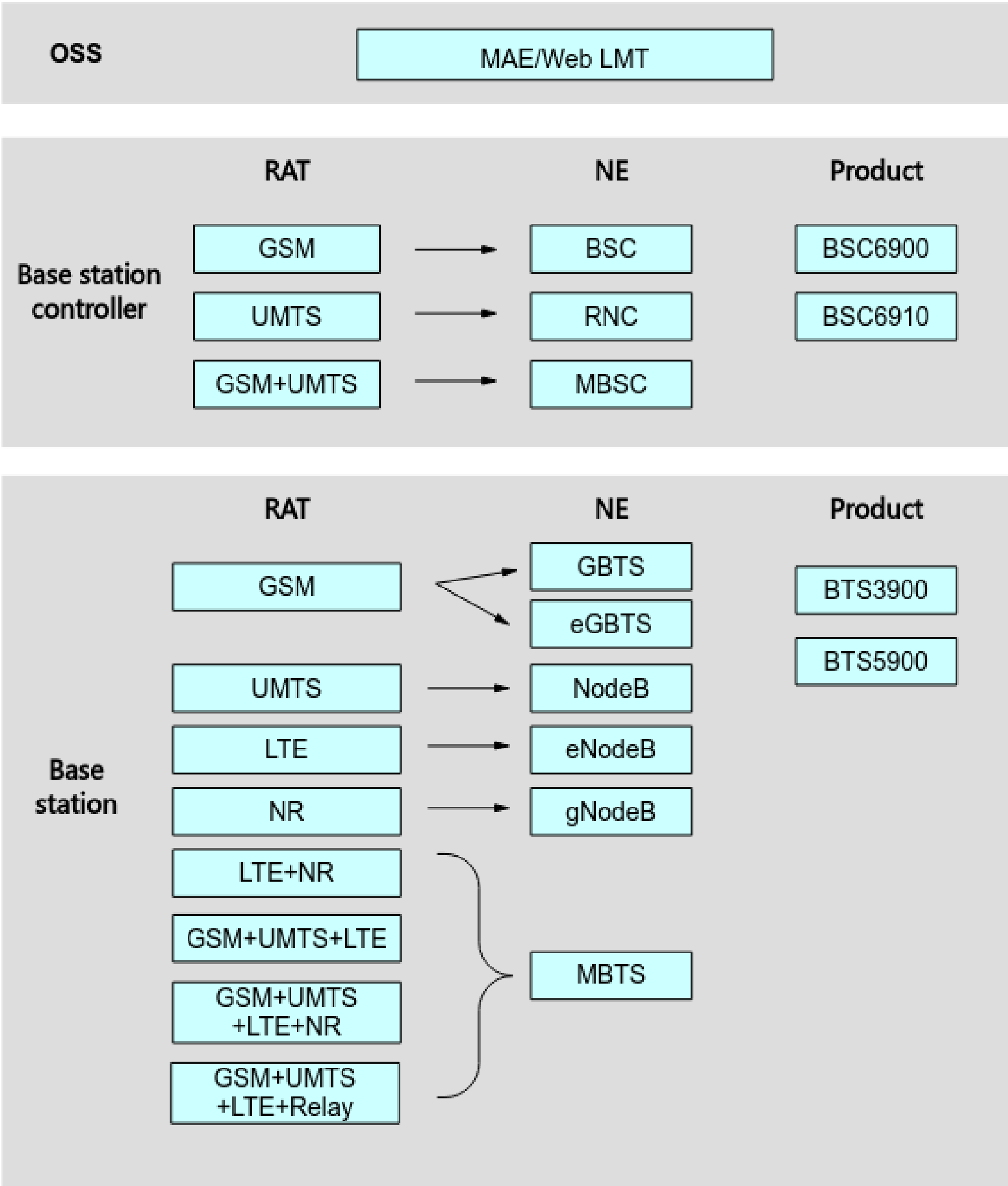
- Transmission solution
- Refarming solution
- Energy saving solution
- Interoperability solution
- Mode evolution solution

For details, see **5 SingleRAN Solution Application**.

4 SingleRAN Solution Overview

Figure 4-1 shows the relationships among typical RATs, NEs, and products on a SingleRAN network.

Figure 4-1 Relationships among typical RATs, NEs, and products on a SingleRAN network



NOTE

The MBTS that supports Relay is applicable to the RuralCow solution. The RAT of Relay is displayed only on the MAE, which facilitates relay O&M of the RuralCow solution. For details about the RuralCow solution, see *RuralCow Solution*.

For ease of description, RATs are abbreviated as follows:

- GSM is abbreviated to G.
- UMTS is abbreviated to U.
- LTE is abbreviated to L. In scenarios where differentiation is required, LTE FDD, LTE TDD, and LTE NB-IoT are used.

- NR is abbreviated to N.

The BBU configuration expressions are normalized as follows in documentation and on GUIs:

- & indicates that the services of different RATs are carried by different main control boards. For example, if a separate-MPT base station is configured with a GSM main control board GTMU_G and an LTE main control board UMPT_L, the RAT deployment of this base station can be described as G&L or GL.
- * indicates that different RATs share the same main control board. For example, if a co-MPT base station is configured with a main control board UMPT_GUL, the RAT deployment of this base station can be described as G*U*L.
- + indicates that two BBUs are used. For example, G&U+L indicates that a GSM main control board and a UMTS main control board are configured in one BBU, and an LTE main control board is configured in the other BBU. For another example, G*U+L indicates that one main control board that processes both GSM and UMTS data is configured in one BBU, and an LTE main control board is configured in the other BBU.

4.1 OSS Products

The MAE manages the following Huawei mobile network equipment:

- RAN equipment
- GBSS equipment
- SingleRAN equipment
- GSM/UMTS core network equipment
- LTE EPC network equipment
- NR 5GC network equipment
- WLAN equipment
- SingleDAS network equipment
- eRGN network equipment
- STP network equipment
- IMS network equipment
- Wireless transport bearer equipment and auxiliary networking equipment used by the mobile network

The MAE-Access provides the following basic network management functions:

- Configuration management
- Performance management
- Fault management
- Security management
- Log management
- Topology management
- Software management

- System management

It also provides a variety of optional functions. The MAE-Access is responsible for the centralized OM in the Huawei mobile NE management solution. The system follows a component-based design idea that the components communicate with each other through the Common Object Request Broker Architecture (CORBA) bus. The system adopts an open architecture, and NEs of different types are accessible through the NE mediation layer. The MAE-Access provides the third-party products or manufacturers with open interfaces, which can be used to interconnect with the equipment of several mainstream vendors.

The MAE-Deployment works as a function component of the MAE data configuration solution. MAE-Deployment basic features are included in MAE basic features, and MAE-Deployment optional features, which are under license control, are used as MAE optional features. The MAE-Deployment implements fast deployment and maintenance of GBSS, RAN, LTE, NR, and SingleRAN networks.

The MAE also has some other products. For example, the MAE-Evaluation is responsible for performance data measurement and service data analysis on a mobile network, thereby implementing network performance management and auxiliary operation.

4.2 Controller Products

Huawei's mainstream controller products are BSC6900 and BSC6910. The use of the BSC6910 is recommended. **Figure 4-2** shows the appearance of the BSC6910.

Huawei's mainstream controller products are BSC6900, BSC6910, and BSC6960. The use of the BSC6910 is recommended. **Figure 4-2** shows the appearance of the BSC6910.

Figure 4-2 BSC6910

The BSC6910 has three variants: BSC6910 GSM, BSC6910 UMTS, and BSC6910 GU.

- The BSC6910 GSM or BSC6910 UMTS is referred to as the BSC6910 in independent mode.
- The BSC6910 GU is referred to as the BSC6910 in integrated mode.

  In integrated mode, GSM and UMTS share physical units, such as the operation and maintenance unit (OMU) and clock processing unit (GCU/GCG). They also share the software of the same version, implementing centralized software management. However, GSM and UMTS service processing boards are separately configured in their own subracks. The BSC6910 in integrated mode must work with a SingleRAN version.

The BSC6910 in independent mode supports co-cabinet deployment. That is, a BSC6910 GSM and a BSC6910 UMTS are deployed in the same cabinet.

The GU integrated mode (with the MPS shared) is not recommended for BSC6910/BSC6900. The separate-cabinet deployment of BSC6910/BSC6900 in independent mode (independent functions of BSC and RNC) is recommended.

Table 4-1 Product variants of a controller

Product Variant	Supported RAT	NE Type	Description
BSC6910 GSM	GSM	BSC6910 GSM	Independent mode
BSC6910 UMTS	UMTS	BSC6910 UMTS	Independent mode
BSC6910 GU	GSM and UMTS	BSC6910 GU	Integrated mode: ● GSM and UMTS share physical units, such as the OMU and GCU/GCG. They also share the software of the same version, implementing centralized software management. ● If a GU dual-mode controller is used, GSM and UMTS controller versions must be upgraded together.
BSC6960 GSM	GSM	BSC6960 GSM	Independent mode
BSC6960 UMTS	UMTS	BSC6960 UMTS	Independent mode

4.3 Base Station Products

Base station products refer to macro and LampSite base stations.

Macro Base Stations

- 3900 series base stations

  3900 series base stations are the mainstream products of macro base stations, and are usually deployed in equipment rooms onsite. These base stations support the GSM, UMTS, LTE, and NR RATs.

  The 3900 series multimode macro base stations adopt a modular design, and consist of cabinets, indoor baseband modules (BBU3900/BBU3910), outdoor baseband modules (BBU3910A/BBU3910C), and RF modules (RFU, RRU, or AAU).

  NOTE

  The BBU3910C supports only GSM, UMTS, LTE, GU, and GL.

  – The BBU3900 and BBU3910 are case-shaped, as shown in **Figure 4-3** and **Figure 4-4**. Each BBU supports the mixed insertion of GSM/UMTS/LTE/NR boards, including the main control boards and baseband processing units.

Some boards such as the UMPT and UBBP support multi-RAT concurrence. From SRAN15.1 onwards, the BBU3910 supports NR (the UMPTe, UMPTg, UMPTga, and UBBPg boards only).

Figure 4-3 BBU3900

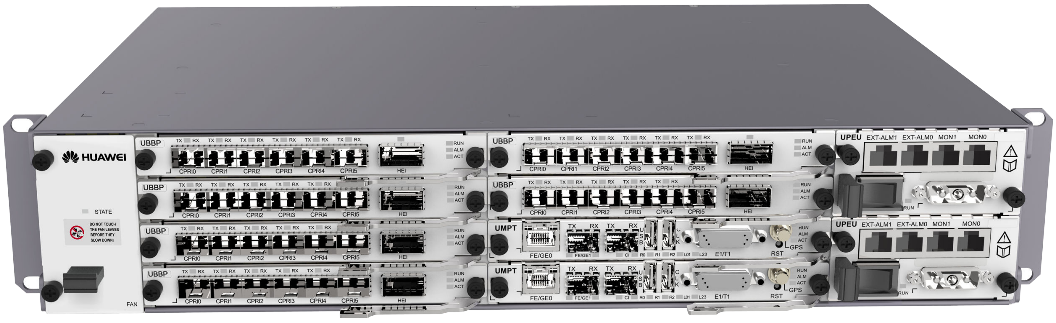
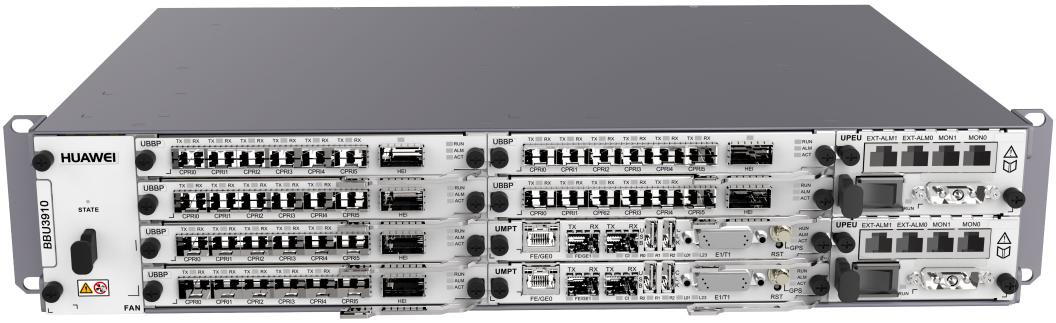


Figure 4-4 BBU3910



– The BBU3910A and BBU3910C adopt the integrated structure, and include the modular mechanical parts and manufactured boards, as shown in **Figure 4-5** and **Figure 4-6**.

Figure 4-5 BBU3910A

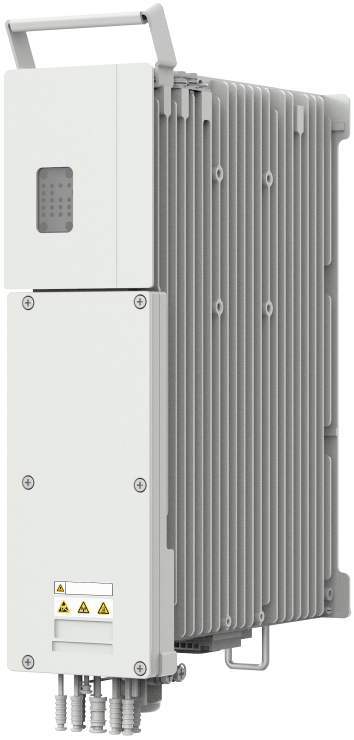


Figure 4-6 BBU3910C

– The baseband processing units and RF modules are connected using electrical cables or optical fibers over CPRI ports. The interface protocols supported by RF modules include CPRI and eCPRI. For details, see *3900 & 5900 Series Base Station Product Documentation*.

Some RF modules, such as the MRFU and RRU3908, have the software-defined radio (SDR) function and support multiple RATs.

- 5900 series base stations

  5900 series base stations are the mainstream products of macro base stations for evolution to future mobile networks, and are usually deployed in equipment rooms onsite. These base stations support the GSM, UMTS, LTE, and NR RATs.

  5900 series multimode macro base stations adopt either of the following architectures:

  These base stations adopt the modular design, and consist of cabinets, indoor baseband units (BBU5900/BBU5910), outdoor baseband units (BBU5900A/BookBBU5901/BBU5900C), and RF modules (RFUs, RRUs, or AAUs).

  These base stations adopt the integrated architecture design and integrate the functions of the BBU, RRU, and relay. The BTS5929R is a typical example.

  – The BBU5900, as shown in **Figure 4-7**, is a next-generation BBU. It differs from the BBU3900 series in exteriors and accommodated board types. In addition to half-width slots, the BBU5900 supports full-width slots. Different slots in the BBU5900 provide different bandwidths in terms of switching capabilities. The BBU5900 can be installed in a Ver.E or later cabinet (a Ver.E cabinet that is reconstructed from a cabinet of another type is also acceptable). The BBU5900 supports the following main control boards and baseband processing units: UMPTb/UMPTe/UMPTg/UMPTga/UBBPd/UBBPe series boards (including UBBPem and UBBPex)/UBBPg series boards and later boards. It does not support the GTMU/UTRP/UBRI boards. The BBU5900 must work with RFUs, RRUs, or AAUs of particular models. For details, see *3900 & 5900 Series Base Station Product Documentation*.

    Figure 4-7 BBU5900

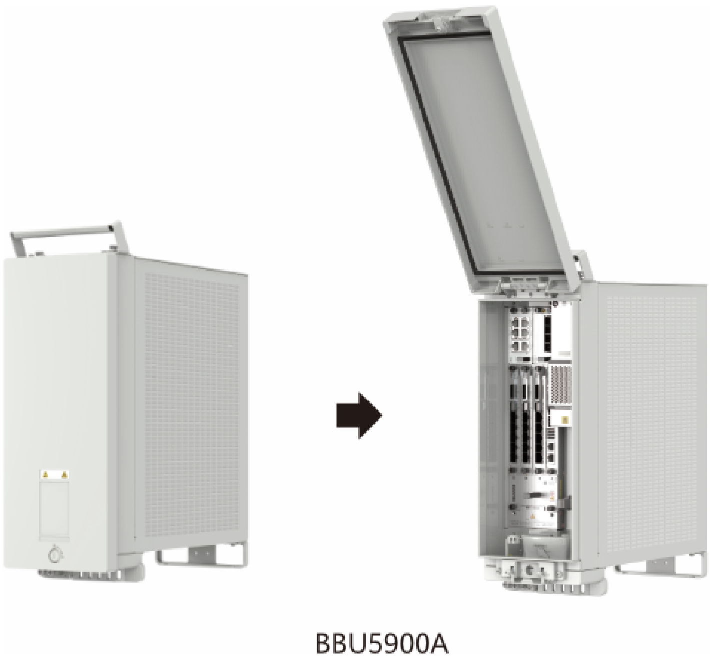
    

  – The specifications of the BBU5910 are the same as those of the BBU5900. **Figure 4-8** shows the appearance of the BBU5910.

Figure 4-8 BBU5910

– The BBU5900A, as shown in **Figure 4-9**, is a new case-shaped outdoor BBU. Each BBU5900A supports the mixed configuration of GSM/UMTS/LTE/NR boards. The slots provided by the BBU5900A are only half of those provided by the BBU5900. The subrack supports the following main control boards and baseband processing units: UMPTe/UMPTg/UMPTga/UBBPd/UBBPe/UBBPf/UBBPg series boards. It does not support the GTMU/UTRP/UBRI/UBBPfw/USCU boards. The BBU5900A must work with RFUs, RRUs, or AAUs of particular models. For details, see *3900 & 5900 Series Base Station Product Documentation*.

Figure 4-9 BBU5900A



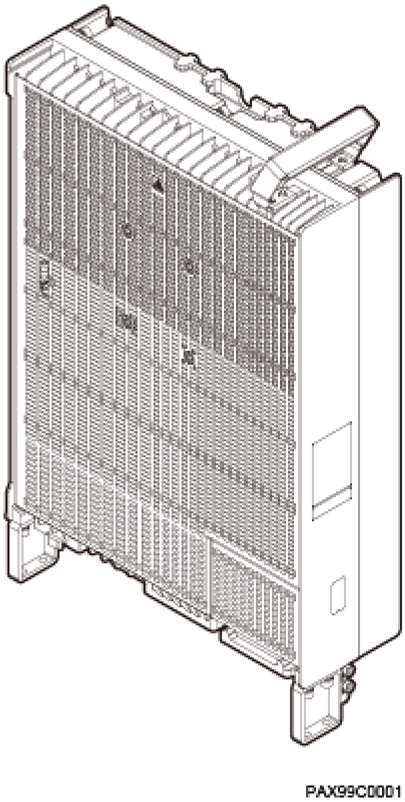
– The BookBBU5901/BBU5900C, as shown in **Figure 4-10** and **Figure 4-11**, respectively, is an outdoor integrated BBU with the functions of main control, transmission, baseband processing, and clock. The BookBBU5901/BBU5900C features small footprint, easy installation, and low power consumption, facilitating quick base station deployment outdoors.

Figure 4-10 BookBBU5901

Figure 4-11 BBU5900C

- The baseband processing units and RF modules are connected using electrical cables or optical fibers over CPRI ports. The interface protocols supported by RF modules include CPRI and eCPRI. For details, see *3900 & 5900 Series Base Station Product Documentation*.
- The BTS5929R is an outdoor integrated base station applicable to rural areas, as shown in **Figure 4-12**. The BTS5929R features easy deployment, simplified O&M, and low power consumption, facilitating quick base station deployment outdoors in rural areas.

Figure 4-12 BTS5929R



The basic modules and auxiliary devices can be flexibly combined into a single-mode base station, a separate-MPT multimode base station, or a co-MPT multimode base station. The flexible combination meets the installation requirements of various sites and satisfies the O&M demands of different RATs. Examples of installation scenarios are indoor and outdoor centralized installation, outdoor distributed installation, and multimode base station installation.

The following part of this document uses 3900 series base stations as an example for illustration.

LampSite Base Stations

The LampSite base stations are distributed base stations designed for indoor coverage. These base stations work in UMTS, LTE, or NR mode and do not work in GSM mode.

4.3.1 Single-Mode Base Station

A single-mode base station is a base station where only one RAT is deployed.

Table 4-2 Product variants of a single-mode base station

Product Variant	Supported RAT	NE Type	Main Control Board
GBTS	GSM	N/A	GTMUc

Product Variant	Supported RAT	NE Type	Main Control Board
eGBTS	GSM	BTS3900 and BTS5900	UMPT/UMDU/MDUC
NodeB	UMTS	BTS3900 WCDMA and BTS5900 WCDMA	UMPT/UMDU/MDUC
eNodeB	LTE	BTS3900 LTE and BTS5900 LTE	UMPT/UMDU
gNodeB	NR	BTS5900 5G and BTS3900 5G	UMPTe/UMPTg/UMPTga

NOTE

The MDUC supports only GSM and UMTS dual-mode.

The GBTS is maintained on the BSC, not on the OSS. From the perspective of O&M, the GBTS is not an independent NE. This type of O&M is called LegacyOM.

The GSM side of a co-MPT multimode base station is an eGBTS, which is maintained directly by the OSS in a way same as that of the NodeB and eNodeB. This type of O&M that is independent of the BSC is called SingleOM.

If the GBTS and co-MPT multimode base station including the eGBTS are both deployed on a network, these two types of GSM base stations are maintained in different ways.

4.3.2 Multimode Base Station

A multimode base station is a base station where two or more RATs are deployed. Multimode base stations are classified into separate-MPT and co-MPT multimode base stations, depending on whether the RATs share the same main control board.

4.3.2.1 Separate-MPT Multimode Base Station

In a separate-MPT multimode base station, multiple RATs are deployed on different main control boards, and correspond to multiple NE types on the OSS.

Two methods of managing common resources are available in a separate-MPT multimode base station: unilateral management and multilateral management.

- Unilateral management: Shared resources are managed by and displayed only on one NE. These resources include the site devices, cabinets, remote electrical tilt (RET) antennas, BBU interconnection modules, transmission resources, clocks, and CPRI links.
- Multilateral management: Common resources are managed by multiple NEs. In this case, operation conflicts must be prevented. For example, in a BBU subrack, the UPEU, UEIU, and FAN are managed by all modes. Multimode RF modules are managed by the modes they serve. The software of the modules

must match the RATs where the base station loading control rights are configured.

From the perspective of the OSS, a separate-MPT multimode base station is a combination of multiple NEs and each NE has an independent O&M channel. The SingleRAN solution requires these single-mode base stations to be managed by the same OSS. In this way, configuration management, alarm management, performance measurement, and software management can be performed collectively.

Table 4-3 Example product variants of a separate-MPT multimode base station

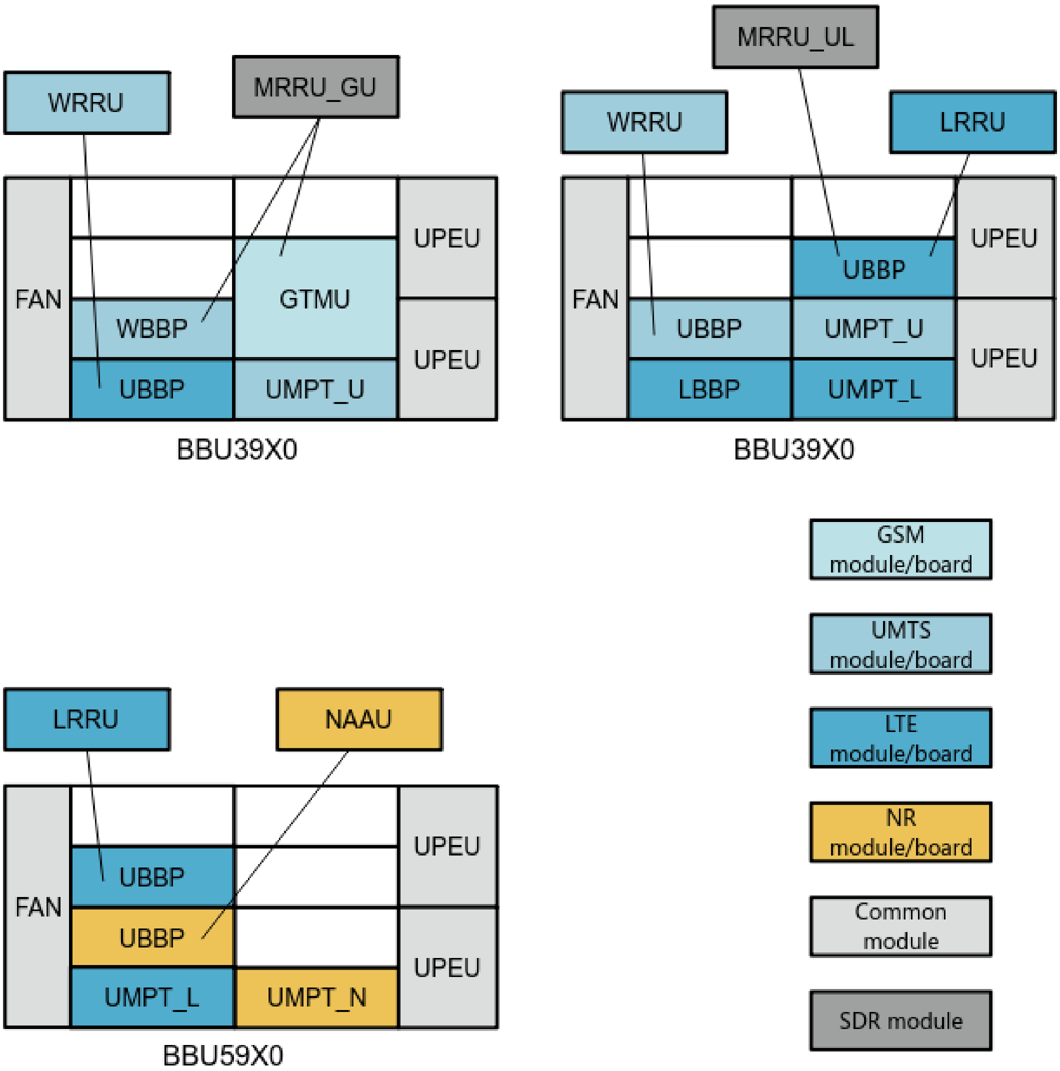
Product Variant	Supported RAT	NE Type	Main Control Board
eGBTS+NodeB	G&U	BTS3900/ BTS5900+BTS3900 WCDMA/BTS5900 WCDMA	UMPT+UMPT
eGBTS +eNodeB	G&L	BTS3900/ BTS5900+BTS3900 LTE/BTS5900 LTE	UMPT+UMPT
NodeB +eNodeB	U&L	BTS3900 WCDMA/ BTS5900 WCDMA +BTS3900 LTE/ BTS5900 LTE	UMPT+UMPT
eNodeB +gNodeB	L&N	BTS3900 LTE/ BTS5900 LTE +BTS3900 5G/ BTS5900 5G	UMPT+UMPTe/ UMPTg
eGBTS+NodeB +eNodeB +gNodeB	G&U+L&N[a]	BTS3900+BTS3900 WCDMA/BTS5900 WCDMA+BTS3900 LTE/BTS5900 LTE +BTS3900 5G/ BTS5900 5G	UMPT+ UMPT +UMPTe/UMPTg
	G*U+L*N[a]	BTS3900/ BTS5900+BTS3900/ BTS5900	UMPT+UMPTe/ UMPTg
	[G*U]&[L*N]	BTS3900/ BTS5900+BTS3900/ BTS5900	UMPT+UMPTe/ UMPTg
a: In separate-MPT scenarios where multimode RRUs are used, the software version of the UMTS main control board must be upgraded to SRAN15.1 or later.			

NOTE

LTE cells can all be configured as FDD or TDD cells according to cell configurations.

Figure 4-13 shows a typical configuration example of a separate-MPT multimode base station.

Figure 4-13 Configuration example of a separate-MPT multimode base station



4.3.2.2 Co-MPT Multimode Base Station

In a co-MPT multimode base station, multiple RATs are deployed on the same main control board, share the same O&M object, and correspond to the same NE on the OSS. In a co-MPT multimode base station, for example, a co-MPT LTE/NR multimode base station, LTE and NR are regarded as two functions of the base station. They share the same O&M channel and correspond to the same BTS3900 or BTS5900 NE on the OSS.

A co-MPT multimode base station adopts the SingleOM mode. One or more RATs can be activated as required on a co-MPT base station. LTE cells can be configured as LTE FDD cells and LTE TDD cells at the same time as required.

Table 4-4 Example product variants of a co-MPT multimode base station

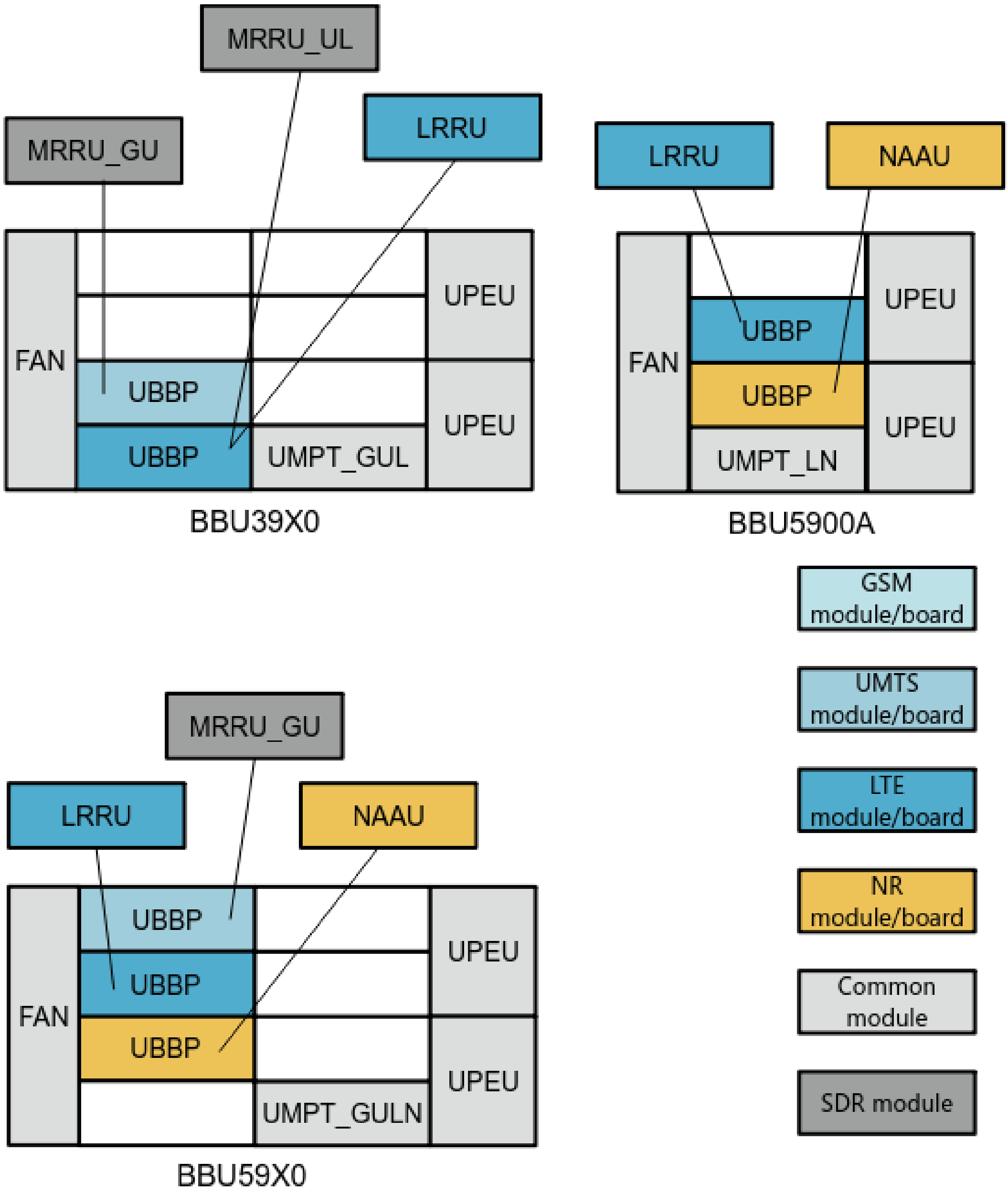
Product Variant	Supported RAT	NE Type	Main Control Board
eGBTS+NodeB	G*U	BTS3900/BTS5900	UMPT/UMDU
NodeB +eNodeB	U*L	BTS3900/BTS5900	UMPT/UMDU
eNodeB +gNodeB	L*N	BTS3900/BTS5900	UMPTe/UMPTg
eGBTS+NodeB +eNodeB +gNodeB	G*U*L*N	BTS3900/BTS5900	UMPTg
eGBTS+NodeB +eNodeB +RNT	G*U*L	BTS5900	UMDU

NOTE

Products with a relay node terminal (RNT) are applicable to the RuralCow solution.

Figure 4-14 shows a typical configuration example of a co-MPT multimode base station.

Figure 4-14 Configuration example of a co-MPT multimode base station



4.4 Other Products

4.4.1 USU3910

The USU3910 is located on the E-UTRAN and is used to interconnect multiple BBUs on a radio network, to implement the Cloud BB solution. The USU3910 supports centralized, distributed, and hybrid BBU interconnection.

4.5 Product Deployment Strategies

4.5.1 Controller Deployment Strategy

The BSC6910 is the recommended controller product.

If a GU dual-mode controller is used, GSM and UMTS controller versions must be upgraded together. That is, the GSM or UMTS controller version cannot be upgraded independently. It is recommended that the BSC6910 work in independent mode. That is, on a SingleRAN network comprised of GSM and UMTS, you are advised to deploy two independent controller products, BSC6910 GSM and BSC6910 UMTS.

4.5.2 Base Station Deployment Strategy

Before deploying a type of base station (single-mode, separate-MPT multimode, or co-MPT multimode), you need to take into consideration the existing base stations, network transition, and O&M mode.

If operators require independent O&M of each mode, single-mode base stations are recommended. In this situation, GSM, UMTS, LTE, and NR networks are not closely related to each other.

If operators want to share the same equipment and O&M to build OneSite, and want each RAT to operate separately, separate-MPT multimode base stations can be used.

If operators want to further share the main control board and make GSM, UMTS, LTE, and NR the four functions of a base station for unified O&M and easy network transition, co-MPT multimode base stations can be used.

On a live network, operators may deploy base stations according to the actual situation. Therefore, there are multiple base station types on the network. All Huawei base stations use SingleOM except the BTS3900 GSM, which uses LegacyOM. Although different base station types are configured with different RATs, their O&M modes and functions are the same.

4.5.3 OSS Deployment Strategy

There are three deployment rules.

Rule 1: A base station controller and its GSM and UMTS base stations must be managed by the same OSS.

Rule 2: All modes of a co-MPT multimode base station must be managed by the same OSS. The reason is that a SingleRAN co-MPT multimode base station is an independent NE, which can only be managed by one OSS.

Rule 3: All modes of a SingleRAN separate-MPT base station must be managed by the same OSS.

4.5.4 Other Deployment Strategies

If coordination is required among E-UTRAN base stations, the USU3910 can be used to interconnect BBUs to implement the Cloud BB solution. This solution provides the following coordination-related features for base stations:

- Coordinated Scheduling based Power Control

- UL CoMP based on Coordinated eNodeB
- Inter-eNodeB CA based on Coordinated eNodeB

5 SingleRAN Solution Application

The SingleRAN network follows the trend of multiple RATs (GULN) and multiple frequency bands. During network evolution, diversified solutions are introduced, including transmission, security, CloudAIR, refarming, energy saving, and interoperability.

5.1 Transmission Solution

Each RAT can use separate transmission or common transmission.

In separate transmission, each mode uses an independent transmission port.

In common transmission, all modes use the same transmission port to access the same transmission network, which implements transmission resource sharing.

Transmission port sharing reduces the number of transmission links. Transmission network sharing simplifies transmission configuration and maintenance. Transmission network adjustments are reduced and smooth network evolution is supported during the evolution from an existing single mode to multiple modes.

The transmission solution involves the features listed in the following table.

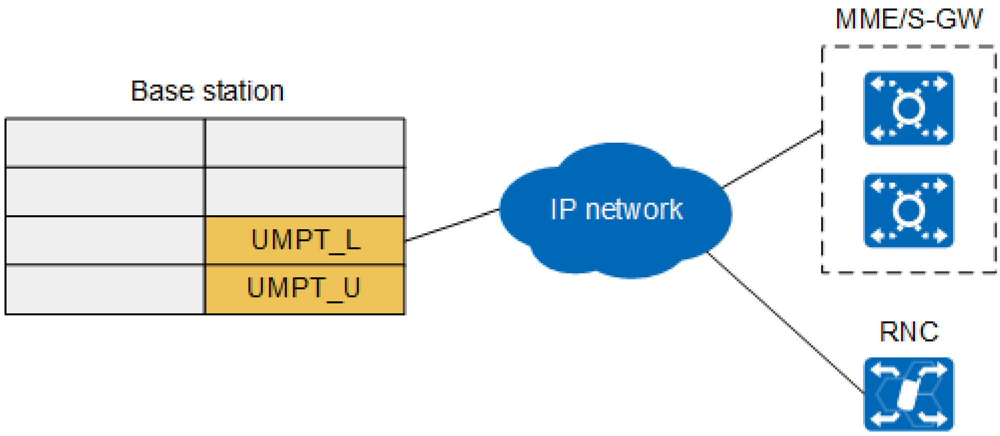
<table>
<tr><th>Feature Name</th><th>Description</th><th>Reference Document</th></tr>
<tr><td>IP-Based BSC and RNC Co-Transmission on MBSC Side</td><td rowspan="2">These features support co-transmission networking on the multimode base station controller side or the co-transmission networking on the multimode base station side. In a co-transmission network, transmission ports and transmission networks are shared.</td><td rowspan="2">Common Transmission</td></tr>
<tr><td>IP-Based Multi-mode Co-Transmission on BS side</td></tr>
</table>

Feature Name	Description	Reference Document
Bandwidth sharing of MBTS Multi-mode Co-Transmission	To enable a co-transmission port to implement unified data scheduling and management, differentiation and fairness among different service types and RATs must be ensured. Moreover, transmission resource congestion when all of the RATs have overlapping traffic bursts must be addressed. Therefore, Huawei introduces the Bandwidth Sharing of Multimode Base Station Co-Transmission feature.	*Bandwidth Sharing of Multimode Base Station Co-Transmission*
Multi-mode BS Common Reference Clock	In the Common Clock feature, all RATs of a multimode base station share a clock source and require only one set of clock equipment.	*Common Clock*
Co-Transmission Resources Management on MBSC	Co-TRM refers to the common management of transmission resources when the IP transmission bandwidth is shared by the GSM and UMTS services of the multimode base station. Co-TRM improves the usage of transmission resources and ensures the transmission quality of services (QoS).	*Common Transmission Resource Management on MBSC* in *GBSS Feature Documentation* or *RAN Feature Documentation*

5.1.1 Separate-MPT Multimode Base Station Co-transmission Through the Backplane

Different main control boards are interconnected through the backplane. The co-transmission networking is shown in **Figure 5-1**.

Figure 5-1 Separate-MPT multimode base station co-transmission through the backplane



This figure uses LTE and UMTS main control boards as an example. The combinations of other modes are similar.

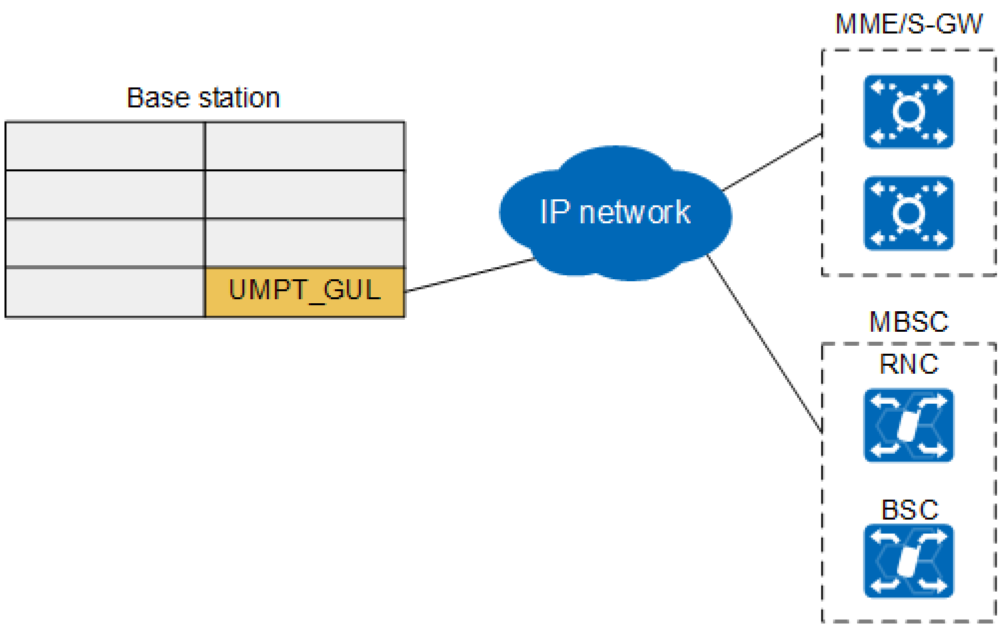
- The LTE main control board is directly connected to the bearer network and performs route forwarding for the UMTS main control board.
- The LTE main control board obtains clock signals from the bearer network and shares the signals with the backplane. The UMTS main control board then obtains clock signals through the backplane.

For details about the configuration, see *Common Transmission*.

5.1.2 Co-MPT Multimode Base Station Co-transmission

The co-transmission networking of a co-MPT multimode base station is shown in **Figure 5-2**.

Figure 5-2 Co-transmission networking of a co-MPT multimode base station



This figure uses the UMPT_GUL as an example. The combinations of other modes are similar.

- The port that connects the UMPT_GUL with the bearer network can carry the services of different modes.
- The UMPT_GUL obtains clock signals from the bearer network and shares the signals with different modes.

For details about the configuration, see *Common Transmission*.

NOTE

The main control board of a co-MPT base station also supports independent transmission of the data of different RATs over different ports.

5.2 Security Solution

To address the increasing severity of network security threats, Huawei has designed and deployed SingleRAN networks to provide multi-plane and multi-layer security protection against various threats and attacks. This design is in accordance with the security model defined in ITU-T X.805.

The security solution involves the features listed in the table below.

Feature Name	Description	Reference Document
OM security	O&M security measures include OMCH security, Web security, user management, personal data security, security management of configuration files, digital signature-based software integrity protection, time security, security alarms/events/logs, OMU anti-attack, security policy level configuration, and security monitoring.	*OM Security*
Equipment security	Equipment security measures include physical security, operating system (OS) security, security environment, self-check upon startup, process auditing, Access Control List (ACL) packet filtering, network attack prevention, and physical port security.	*Equipment Security*

Feature Name	Description	Reference Document
IPsec	Huawei base station controllers and base stations support IPsec networking, which ensures data confidentiality, integrity, and authentication.	*IPsec*
PKI	PKI binds a certificate holder's identity with a public key using signed digital certificates to facilitate the acquisition, access to, and revocation of certificates. Additionally, PKI uses digital certificates and related services to authenticate the identities of communicating entities, which ensures confidentiality, integrity, and non-repudiation of communication data. Huawei base stations support initial certificates and comply with Certificate Management Protocol version 2 (CMPv2).	*PKI*
Access Control based on 802.1x	Huawei base stations can access the customer network using 802.1x-based access control, which authenticates equipment on the access port. Only service data that passes authentication can be transmitted through the port. This prevents unauthorized equipment from accessing the customer network.	*Access Control based on 802.1x*
SSL	SSL is a protocol that provides end-to-end communication security between the Transmission Control Protocol (TCP) layer and the application layer.	*SSL*
MACsec	MACsec is used to protect data transmitted between interconnected UMPT boards.	*MACsec*

5.3 CloudAIR Solution

The CloudAIR feature package contains features related to spectrum sharing, channel sharing, and power sharing.

CloudAIR Spectrum Sharing

CloudAIR spectrum sharing features implement spectrum sharing between LTE FDD and NR, GSM and UMTS, GSM and LTE, and UMTS and LTE.

Function	Description	Reference Document
GSM and UMTS spectrum sharing	GSM and UMTS spectrum sharing allows GSM and UMTS networks to be co-deployed on a minimum spectrum of 5 MHz in single-UMTS carrier scenarios, and on a minimum spectrum of 8.8 MHz in dual-UMTS-carrier scenarios. It introduces a joint scheduling technology to reduce the interference that uplink and downlink UMTS signals produce on the GSM network.	*GU@5MHz Joint Scheduling* in *GBSS Feature Documentation* *GU@5 MHz* in *RAN Feature Documentation* *1.2 MHz Networking for BCCH TRXs* in *GBSS Feature Documentation* *DC-HSDPA Shared with GSM* in *RAN Feature Documentation*
GSM and LTE spectrum sharing	GSM and LTE spectrum sharing enables quick LTE deployment using the existing GSM spectrum, or enables an existing LTE network to share a part of GSM spectrum. In either case, this feature allows GSM and LTE to share GSM spectrum based on traffic volume.	*GSM and LTE Spectrum Concurrency* in *GBSS Feature Documentation* or *eRAN Feature Documentation*

Function	Description	Reference Document
UMTS and LTE Spectrum Sharing	UMTS and LTE Spectrum Sharing enables UMTS and LTE cells to work on a part of continuous spectrum. The UMTS and LTE networks collaboratively allocate the shared spectrum resources based on the LTE traffic volume. This ensures the UMTS services and at the same time increases the LTE bandwidth.	*UMTS and LTE Spectrum Sharing* in *RAN Feature Documentation* or *eRAN Feature Documentation* *UMTS and LTE Spectrum Sharing Based on DC-HSDPA* in *RAN Feature Documentation* or *eRAN Feature Documentation*
LTE FDD and NR Uplink Spectrum Sharing	LTE FDD and NR Uplink Spectrum Sharing allows LTE to share a part of uplink low-frequency spectrum with NR. After LTE FDD and NR Uplink Spectrum Sharing is implemented, UL and DL Decoupling or Super Uplink is used to establish the supplementary uplink (SUL) cell using the shared uplink low-frequency spectrum for NR. This improves the NR uplink coverage (when UL and DL Decoupling is used) or the uplink throughput (when Super Uplink is used).	*LTE FDD and NR SUL Dynamic Spectrum Sharing* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*
LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE FDD and NR Flash Dynamic Spectrum Sharing enables LTE FDD and NR to dynamically share time-frequency resources on a spectrum segment based on the traffic volumes of LTE and NR networks. In the time domain, instantaneous spectrum sharing is supported on a per 1 ms basis. In the frequency domain, dynamic spectrum sharing is performed per RB. During spectrum sharing, uplink and downlink physical channel resources are coordinately scheduled to prevent interference between LTE FDD and NR.	*LTE FDD and NR Dynamic Spectrum Sharing* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*

CloudAIR Power Sharing

CloudAIR power sharing features implement dynamic power sharing between LTE TDD and NR, between UMTS and LTE, and between LTE carriers.

Function	Description	Reference Document
UMTS and LTE Dynamic Power Sharing	When the power usage of UMTS cells is low, UMTS cells can share the idle power with LTE cells. When the power usage of LTE cells is low, LTE cells can share the idle power with UMTS cells.	*UMTS and LTE Dynamic Power Sharing* in *RAN Feature Documentation* or *eRAN Feature Documentation*
Dynamic Power Sharing Between LTE Carriers	Dynamic Power Sharing Between LTE Carriers enables multiple LTE carriers on the same RF channel to share power. The bandwidth of an LTE carrier, in a given instant, can be fully occupied or become idle. If further data scheduling is required on a carrier where all its bandwidth is already occupied, this carrier can borrow power from idle carriers. The instantaneous power available for the first carrier may exceed its statically configured power.	*Dynamic Power Sharing Between LTE Carriers* in *eRAN Feature Documentation*

Function	Description	Reference Document
LTE TDD and NR Flash Dynamic Power Sharing	LTE and NR services are bursty, and the power of LTE and NR carriers, in a given instant, can be fully occupied or become idle. LTE and NR Flash Dynamic Power Sharing enables 1-ms-level dynamic power sharing between LTE TDD and NR carriers that share the same RF channel of an RF module. The unused power of the instantaneous idle carriers of one RAT can be used by the carriers of the other RAT if the bandwidth of the latter RAT is already occupied and further data scheduling is still required, so that the instantaneous power used by a carrier exceeds its static power configuration. LTE and NR power sharing is allowed based on the fact that LTE TDD and NR carriers do not have their power fully occupied at the same time, and helps improve power efficiency.	*LTE and NR Power Allocation* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*

CloudAIR Channel Sharing

CloudAIR channel sharing consists of LTE spectrum coordination and UL and DL Decoupling.

Function	Description	Reference Document
LTE Spectrum Coordination	LTE spectrum coordination improves the downlink usage of CA UEs at the uplink coverage edge of high frequencies.	*LTE Spectrum Coordination* in *eRAN Feature Documentation*
UL and DL Decoupling	UL and DL Decoupling enables downlink data to be transmitted in the C-band and uplink data to be transmitted in a sub-3 GHz band (for example, 1.8 GHz), thereby improving uplink coverage.	*UL and DL Decoupling* in *5G RAN Feature Documentation*

5.4 Refarming Solution

The refarming solution enables operators to reuse frequency resources and introduce a new radio communication technology to improve spectral efficiency and data traffic volume. After spectrum refarming, the cells of two different modes work in the same band. To reduce costs, SDR modules are generally used in the cells where refarming is performed, which makes power sharing possible between the cells.

5.4.1 Frequency Sharing

GSM spectrums are golden ones. With the development of the network, GSM users gradually migrate to the UMTS or LTE network. As a result, GSM traffic volume and spectrum efficiency are becoming increasingly low. To fully utilize GSM spectrums, the operator refarms GSM spectrums to UMTS and LTE in most cases, which correspond to GU refarming and GL refarming respectively. The following table provides some details. Spectrum sharing requires that the GSM, UMTS, and LTE network equipment be provided by Huawei.

Table 5-1 Functions involved in the frequency sharing solution

Function	Description	Reference Document
GU refarming	GU refarming enables GSM to allocate a bandwidth of about 5 MHz to UMTS. The key technologies include UMTS flexible bandwidth and narrowband interference suppression.	*GU 900 MHz Non-standard Frequency Spacing* in *GBSS Feature Documentation* or *RAN Feature Documentation*
GL refarming	GL refarming enables GSM to allocate a certain bandwidth to LTE. Most operators first allocate the GSM 1800 MHz band, whereas some operators first allocate the GSM 900 MHz band. The key technique is LTE non-standard bandwidth.	*Compact Bandwidth (FDD)* in *eRAN Feature Documentation*

5.4.2 Power Sharing

After GSM spectrums are used by UMTS or LTE through refarming, GSM power can be shared with the UMTS or LTE cell when GSM traffic is low on the condition that the GU900, GL900, and GL1800 cells share the same SDR RF module. This is to improve the throughput of the UMTS cell and the cell edge users' throughput of the LTE cell. The following table provides some details. Power sharing requires that the GSM, UMTS, and LTE network equipment be provided by Huawei.

Table 5-2 Functions involved in the power sharing solution

Function	Description	Reference Document
GSM and UMTS power sharing	When GSM and UMTS use the SDR RF module, GSM shares the remaining power with UMTS to improve the throughput of the UMTS cell if GSM traffic is low.	*GSM and UMTS Dynamic Power Sharing* in *GBSS Feature Documentation* or *RAN Feature Documentation*
GSM and LTE power sharing	When GSM and LTE use the SDR RF module, GSM shares the remaining power with LTE to improve the cell edge users' throughput of the LTE cell if GSM traffic is low.	*GSM and LTE Dynamic Power Sharing* in *GBSS Feature Documentation* or *eRAN Feature Documentation*

5.5 Energy Saving Solution

Operators deploy multiple frequency bands and RATs in an area to meet the requirements of exponentially growing wireless services. As a result, the network becomes increasingly complex, and the overall energy consumption of devices remains high. Multi-band and multi-RAT coordinated energy saving is the key to energy saving and emission reduction, including multi-RAT symbol power saving, multi-RAT channel shutdown, and multi-RAT carrier shutdown. The multi-band and multi-RAT coordinated energy saving solution requires that base stations of all RATs be provided by Huawei.

Multi-RAT Symbol Power Saving

Table 5-3 Multi-RAT symbol power saving solution

Function	Description	Reference Document
LTE and NR coordinated symbol power saving	LTE and NR carriers are deployed on the same RF unit. When no data is carried in LTE and NR symbols, the PA can be dynamically shut down to save energy and reduce emission.	*Multi-RAT Coordinated Energy Saving*

Multi-RAT Channel Shutdown

Table 5-4 Multi-RAT channel shutdown solutions

Function	Description	Reference Document
UL coordinated channel shutdown	UMTS and LTE carriers can be deployed on the same RF unit. During off-peak hours, some RF channels can be shut down to save energy and reduce emission.	*Multi-RAT Coordinated Energy Saving*
LNR coordinated channel shutdown	LTE TDD and NR TDD carriers can be deployed on the same RF unit. During off-peak hours, some RF channels can be shut down to save energy and reduce emission.	*Multi-RAT Coordinated Energy Saving*

Multi-RAT Carrier Shutdown

Table 5-5 Multi-RAT carrier shutdown solutions

Function	Description	Reference Document
GSM and UMTS carrier joint shutdown	When a GSM cell and a UMTS cell cover the same area and the UMTS traffic volume is low, UEs in the UMTS cell are handed over to the GSM cell and then the UMTS cell is shut down.	*Multi-RAT Coordinated Energy Saving*
GSM and LTE carrier joint shutdown	When a GSM cell and an LTE cell cover the same area and the LTE traffic volume is low, UEs in the LTE cell are handed over to the GSM cell and then the LTE cell is shut down.	

Function	Description	Reference Document
UMTS and LTE carrier joint shutdown	When a UMTS cell and an LTE cell cover the same area and the LTE traffic volume is low, UEs in the LTE cell are handed over to the UMTS cell and then the LTE cell is shut down.	
GSM, UMTS, and LTE carrier joint shutdown	When GSM, UMTS, and LTE cells cover the same area and the LTE traffic volume is low, UEs in the LTE cell are handed over to the UMTS cell and then the LTE cell is shut down. If the traffic volume in the UMTS cell is also low, UEs in the UMTS cell are handed over to the GSM cell and then the UMTS cell is shut down.	
LTE and NR carrier joint shutdown	If an LTE cell and an NR cell cover the same area and the NR traffic volume is low, UEs in the NR cell are handed over to the LTE cell and then the NR cell is shut down.	*Multi-RAT Coordinated Energy Saving*

5.6 Interoperability Solution

The interoperability solution is to ensure the optimal user experience. It covers the functions related to camping and reselection, voice solutions, service continuity, service steering, and load balancing.

Table 5-6 Functions involved in the interoperability solution

Function	Description	Reference Document
Camping and reselection	The main purpose of cell reselection is to make UEs camp on a suitable network, which implements load balancing between networks and ensures optimal user experience. There are two ways of cell reselection: cell rank-based and priority-based.	GBSS feature parameter description documents: ● *Interoperability Between GSM and WCDMA* ● *Interoperability Between GSM and LTE* RAN feature parameter description documents: ● *Interoperability Between UMTS and LTE* ● *UE Behaviors in Idle Mode* *Idle Mode Management* in *eRAN Feature Documentation* *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*

<table>
<tr><th>Function</th><th>Description</th><th>Reference Document</th></tr>
<tr><td>Voice solutions</td><td>The LTE-related voice solutions include the following: CSFB, SRVCC, and fast return to LTE.
The NR-related voice solutions include the following: EPS fallback, emergency voice fallback, and fast return to NR.</td><td>eRAN feature parameter description documents:
● CS Fallback
● SRVCC
RAN feature parameter description documents:
● Fast CS Fallback Based on RIM
● Interoperability Between UMTS and LTE
Fast LTE Reselection at 2G CS Call Release in GBSS Feature Documentation
Interoperability Between E-UTRAN and NG-RAN in Idle Mode and Interoperability Between E-UTRAN and NG-RAN in Connected Mode in eRAN Feature Documentation or 5G RAN Feature Documentation</td></tr>
</table>

Function	Description	Reference Document
Service continuity	The main purpose of service continuity is to hand over UEs at the edge of a cell in a certain mode to a cell with good signal quality in another mode, which ensures users' service continuity.	GBSS feature parameter description documents: ● *Interoperability Between GSM and WCDMA* ● *Interoperability Between GSM and LTE* RAN feature parameter description documents: ● *Handover* ● *Interoperability Between UMTS and LTE* *Mobility Management in Connected Mode* in *eRAN Feature Documentation* *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*
Service steering	Service steering enables UEs to be handed over from an LTE cell to an NR cell based on the QCI policy. UEs can have better user experience in the NR cell.	*Interoperability Between E-UTRAN and NG-RAN in Idle Mode* and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*

Function	Description	Reference Document
Load balancing	The main purpose of load balancing is to migrate the load of a high-load cell in a certain RAT to a cell in another RAT, which balances the load among GSM, UMTS, and LTE and improves network traffic volume and user throughput.	GBSS feature parameter description documents: • *Interoperability Between GSM and WCDMA* • *Interoperability Between GSM and LTE* *Load Reshuffling* in *RAN Feature Documentation* *Inter-RAT Mobility Load Balancing* in *eRAN Feature Documentation*

5.7 Mode Evolution Solution

Mode evolution of a multimode base station affects services on the live network. Therefore, it is good practice to make preparations and perform the operations that do not affect services in advance, and perform the operations that affect services in parallel. This can shorten the service interruption duration.

Activating configuration data and adjusting hardware are two key operations for mode evolution. They interrupt services, are interdependent of each other, and should preferably be performed in parallel to minimize the service interruption duration. Only after these two operations are completed and when no configuration conflict occurs can services be restored. For example, when a CPRI port on the BBU is adjusted during evolution, data related to the CPRI link must be adjusted accordingly. Services of a multimode base station can be restored only when the port configuration is consistent with the CPRI link configuration. **Figure 5-3** shows the mode evolution procedure.

Figure 5-3 Mode evolution procedure

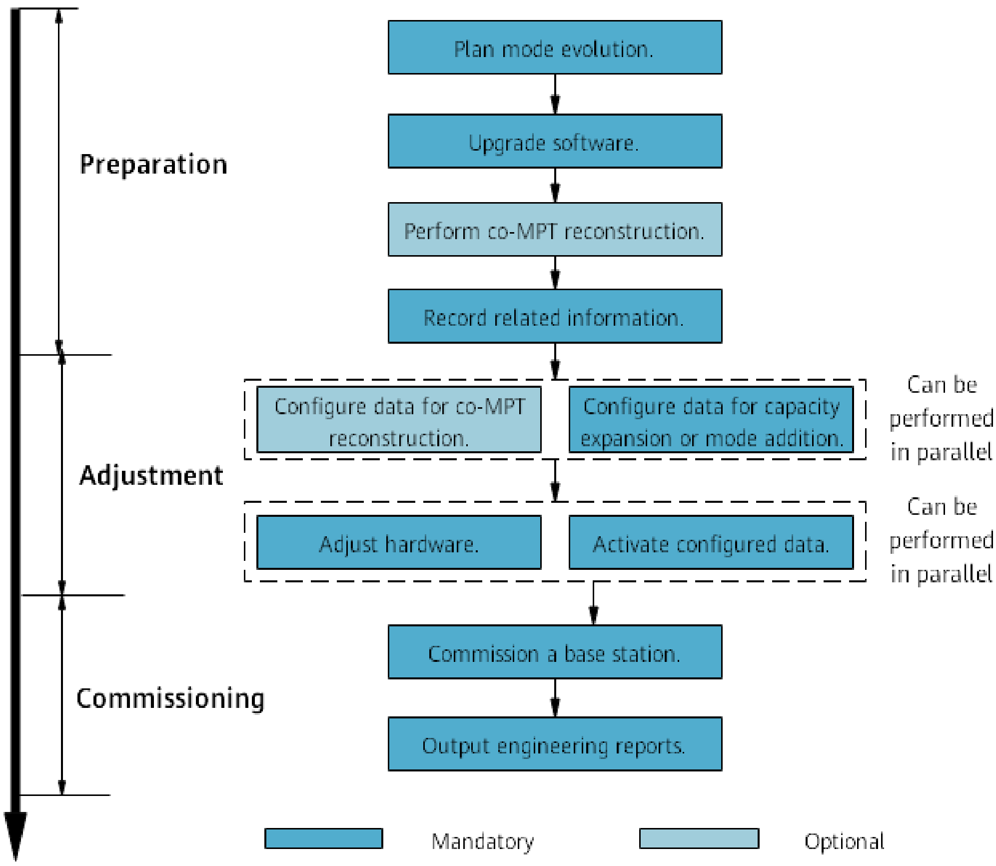


Table 5-7 Procedure for mode evolution of multimode base stations

Phase	Item	Description
Preparation	Plan mode evolution.	1. Make a data plan before and after evolution on power consumption after mode evolution, type and number of each board, cable connections, transmission scheme, CPRI topology, and reference clock. 2. Use the cabling tool on the MAE-Deployment to analyze the RF modules, slots, and CPRI connections. 3. Prepare hardware. 4. Prepare software and license files. 5. Use the MAE-Deployment to make a data configuration file required after mode evolution.
	Upgrade software.	Skip this step if the base station uses the software of the target version.

Phase	Item	Description
	(Optional) Perform co-MPT reconstruction.	Perform co-MPT reconstruction, which is optional, before the evolution from a single mode to multiple modes.
	Record related information.	Record the active alarms, cell status, service KPIs for comparison before and after mode evolution.
Implementation	(Optional) Configure data for co-MPT reconstruction.	In the evolution from a single-mode base station to a co-MPT multimode base station, co-MPT reconstruction must be first performed. Configure data using the MAE-Deployment according to the data plan.
	Configure data for capacity expansion or mode addition.	Configure data using the MAE-Deployment according to the data plan.
	Adjust hardware.	Perform these two operations in parallel to shorten the service interruption duration.
	Activate configured data.	
Verification	Commission a base station.	For the mode that has not been deployed before the mode evolution, you are advised to commission a new mode as a newly deployed base station because the commissioning methods are the same. For the mode that has been deployed before the mode evolution, commission the mode in either of the following ways: ● Check engineering quality and conduct a dialing test if no major changes are made to the main control board and transmission scheme. ● Commission the mode as in new deployment scenarios if major changes are made to the main control board or transmission scheme.
	Output engineering reports.	Summarize the swapping and make engineering reports.

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.